BEFORE YOU READ

Eight kinds of event

Events are subsets of the sample space. Some are defined by what they contain; others are built from other events.

- L0 **Types of events**
  - L1 **Defined by what they contain**
    - L2 **Elementary** exactly one outcome — $\{\omega\}$
    - L2 **Empty** cannot happen — $\emptyset$
    - L2 **Certain** always happens — Ω
  - L1 **Built from other events**
    - L2 **Complement** everything outside the event — A^c
    - L2 **Union** at least one occurs ("or") — $A \cup B$
    - L2 **Intersection** both occur ("and") — $A \cap B$
    - L2 **Difference** in A but not in B — $A - B$
  - L1 **A relation between events**
    - L2 **Disjoint** cannot occur together — $A \cap B = \emptyset$

TAKEAWAY

Events are sets

Every kind of event is a set, so the set operations give the vocabulary: complement, union, intersection and difference.